

Imperfections in Solids

There is no such thing as a perfect crystal.

- What are these imperfections?
- Why are they important?

Many of the important properties of materials are due to the presence of imperfections.



An ideal crystal can be described in terms a three-dimensionally periodic arrangement of points called lattice and an atom or group of atoms associated with each lattice point called unit cell:

$$\text{Crystal} = \text{Lattice} + \text{Unit Cell}$$

However, there can be deviations from this ideality.

These deviations are known as crystal defects.

What are lattice points?

A lattice point is a point at the intersection of two or more grid lines in a regularly spaced array of points, which is a point lattice. In a plane, point lattices can be constructed having unit cells in the shape of a square, rectangle, hexagon, and other shapes.



Types of Imperfections

- Vacancy atoms
- Interstitial atoms
- Substitutional atoms

Point defects

- Dislocations

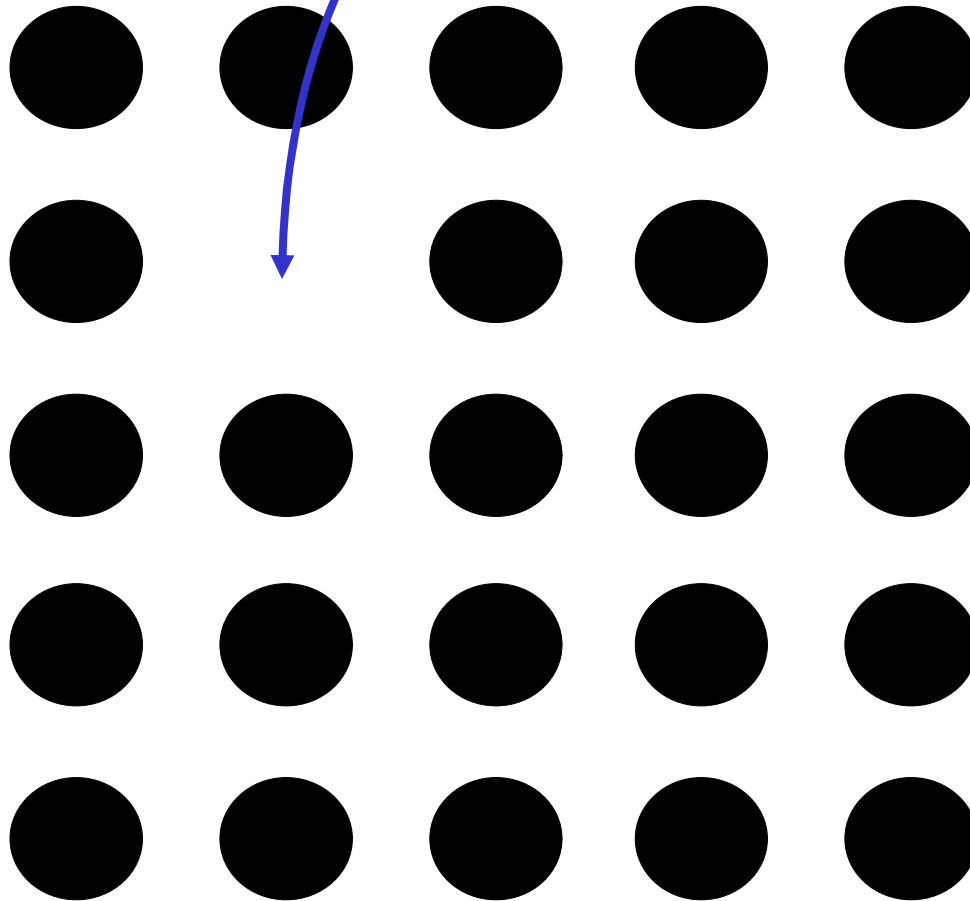
Line defects

- Grain Boundaries

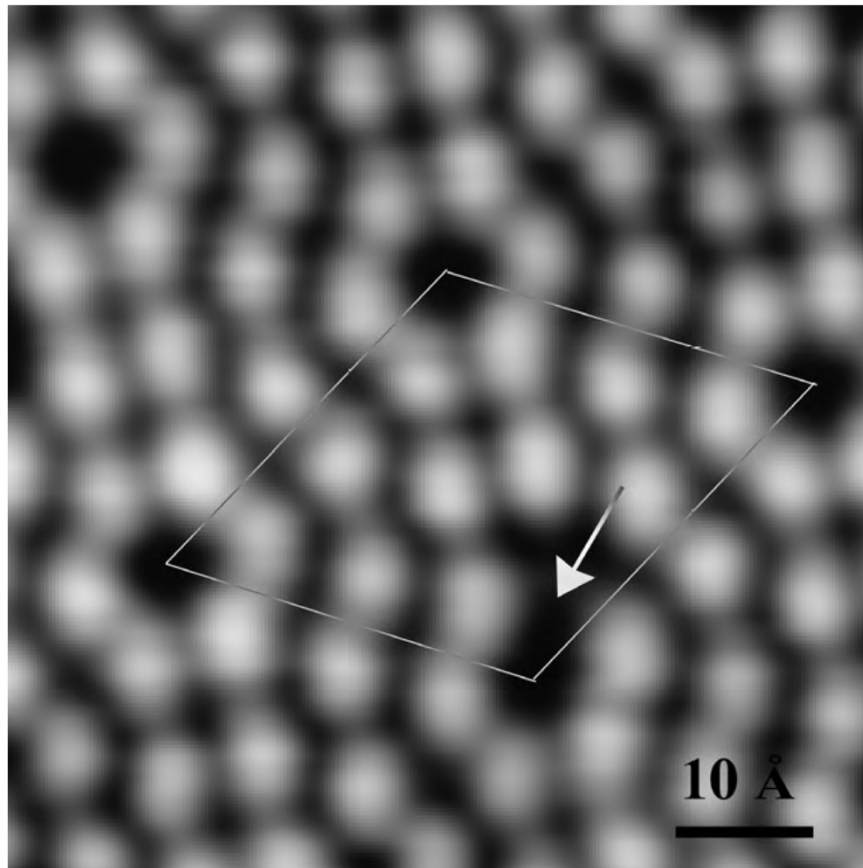
Area defects



Vacancy: A point defect



Vacancy



scanning probe micrograph (generated using a scanning-tunneling microscope) that shows a (111)-type surface plane* for silicon. The arrow points to the location of a silicon atom that was removed using a tungsten nanotip probe. This site from which an atom is missing is the surface analogue of a vacancy defect—that is, a vacant lattice site within the bulk material. Approximately 20,000,000. (Micrograph courtesy of D. Huang, Stanford University.)

Crystalline Imperfections

- “Crystalline defect” is a lattice irregularity having one or more of its dimensions on the order of an atomic diameter.
- Classification of crystalline imperfections is frequently made according to geometry or dimensionality of the defect.



<i>Defects</i>	<i>Dimensionality</i>	<i>Examples</i>
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Point	0	Vacancy
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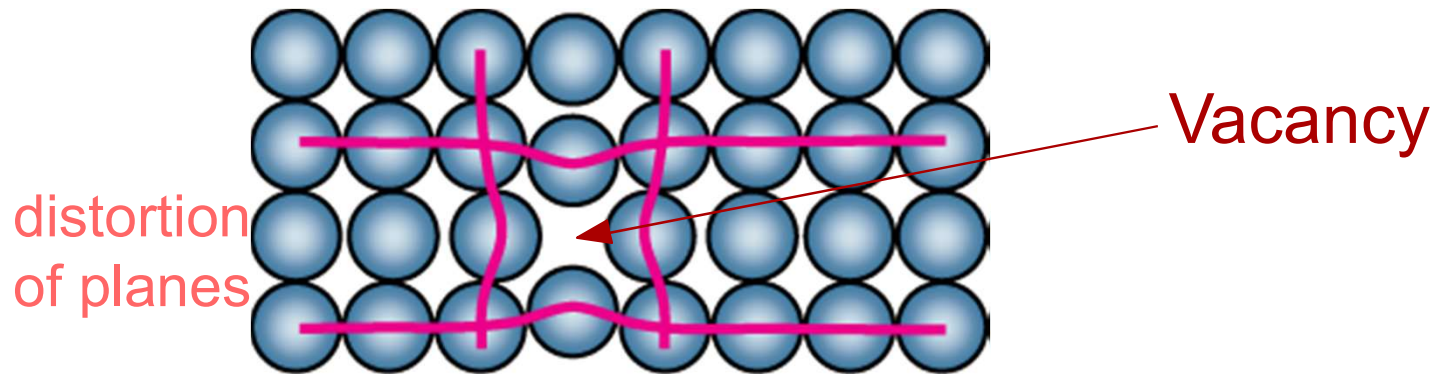
Line	1	Dislocation
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Surface	2	Free surface, Grain boundary
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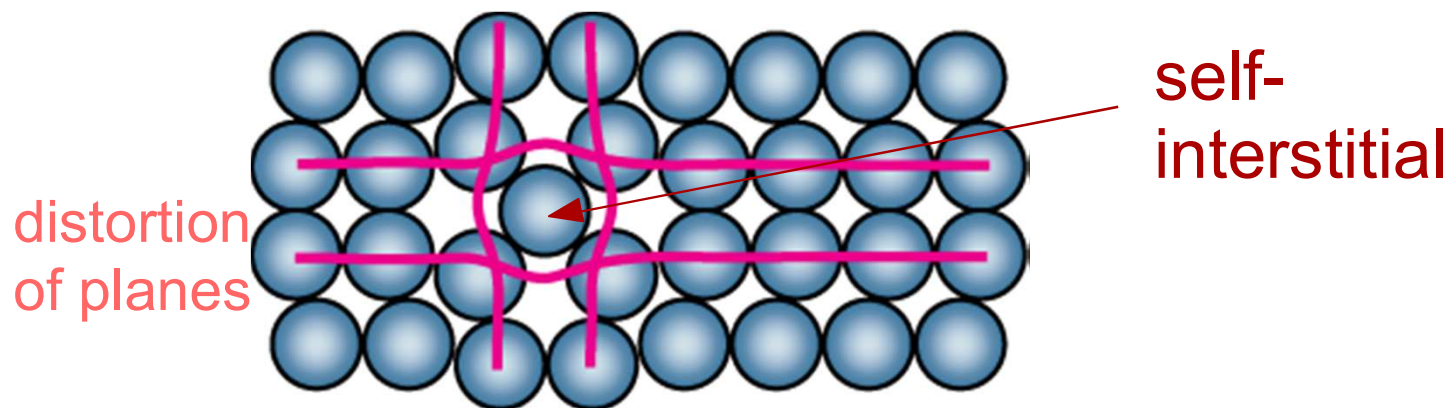


Point Defects

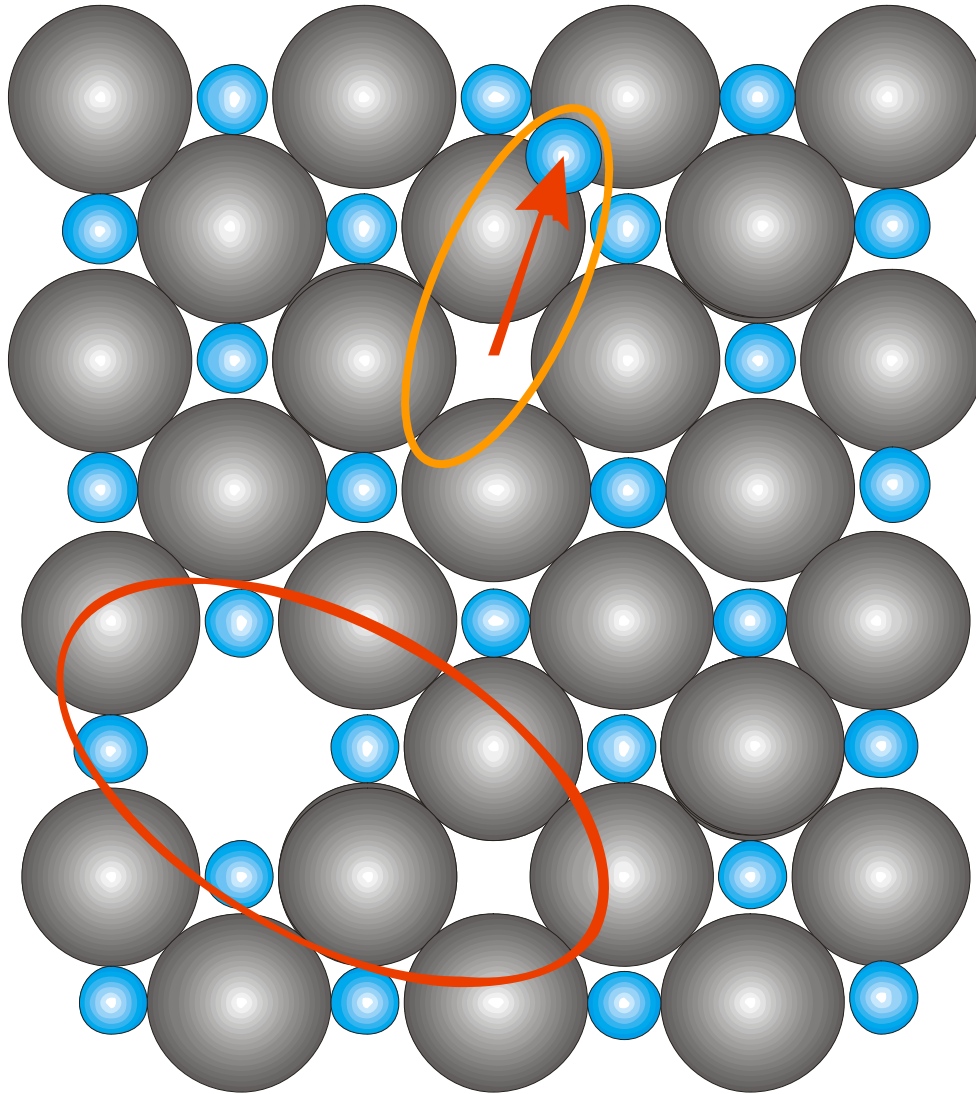
- **Vacancies:**
-vacant atomic sites in a structure.



- **Self-Interstitials:**
-"extra" atoms positioned between atomic sites.



Defects in Ionic Solids



Frenkel
defect

Cation vacancy
+
cation interstitial

Schottky
defect

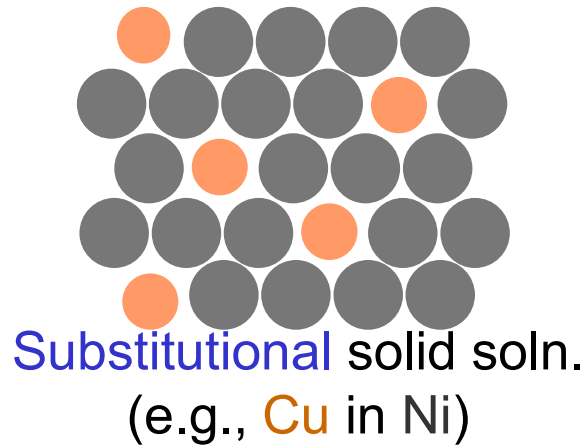
Cation vacancy
+
anion vacancy



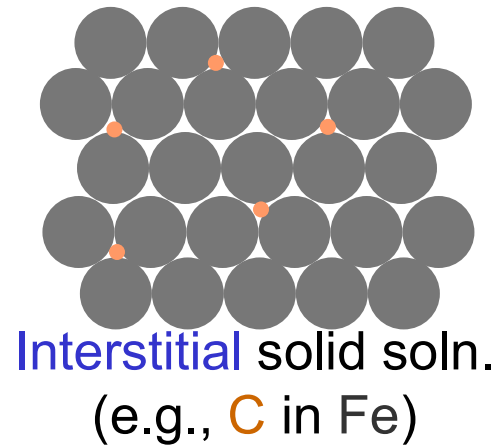
Impurities in Solids

Two outcomes if impurity (B) added to host (A):

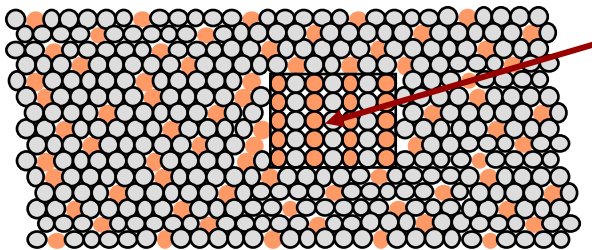
- Solid solution of B in A (i.e., random dist. of point defects)



OR



- Solid solution of B in A plus particles of a new phase (usually for a larger amount of B)



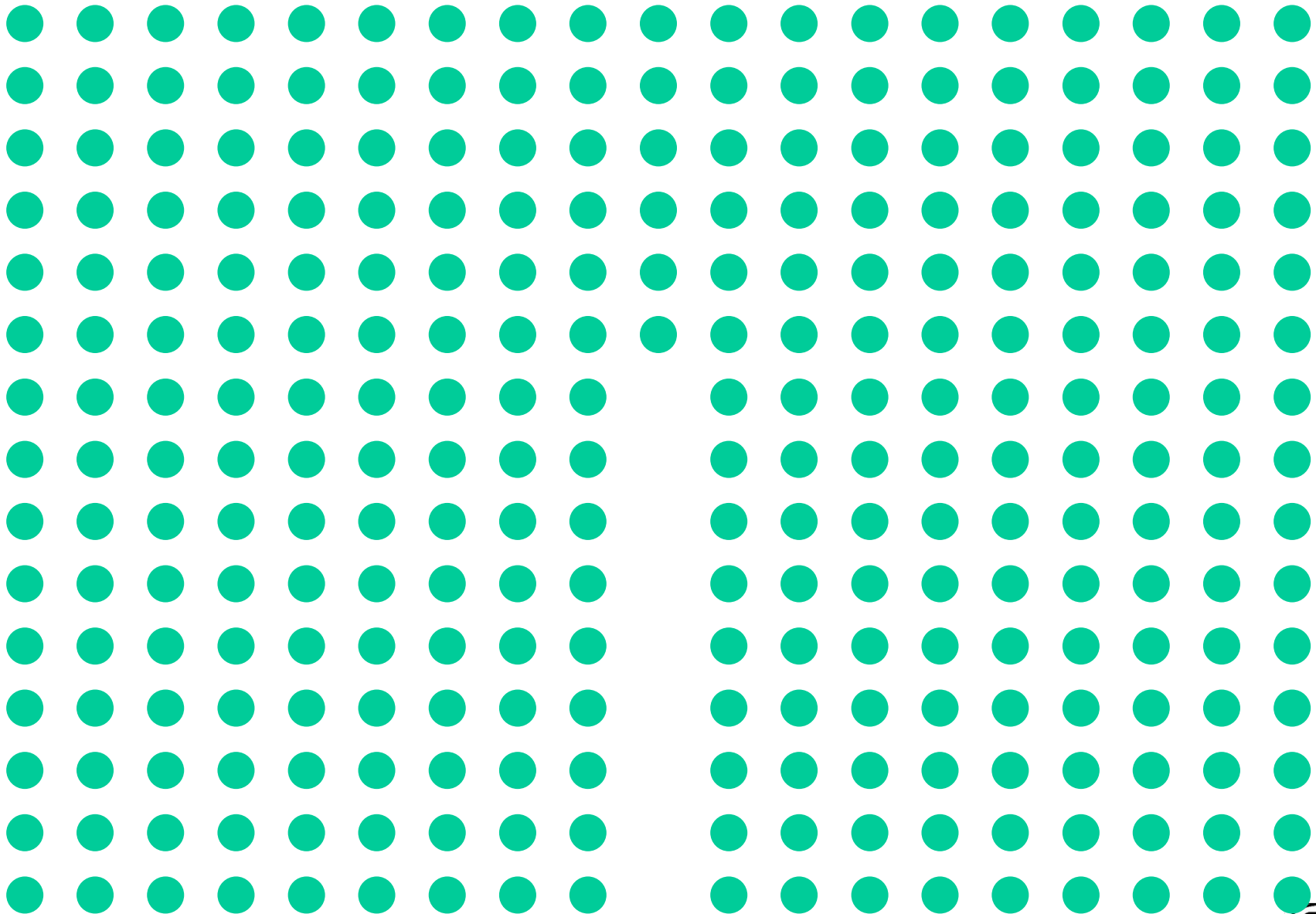
Second phase particle
--different composition
--often different structure.

Line Defects

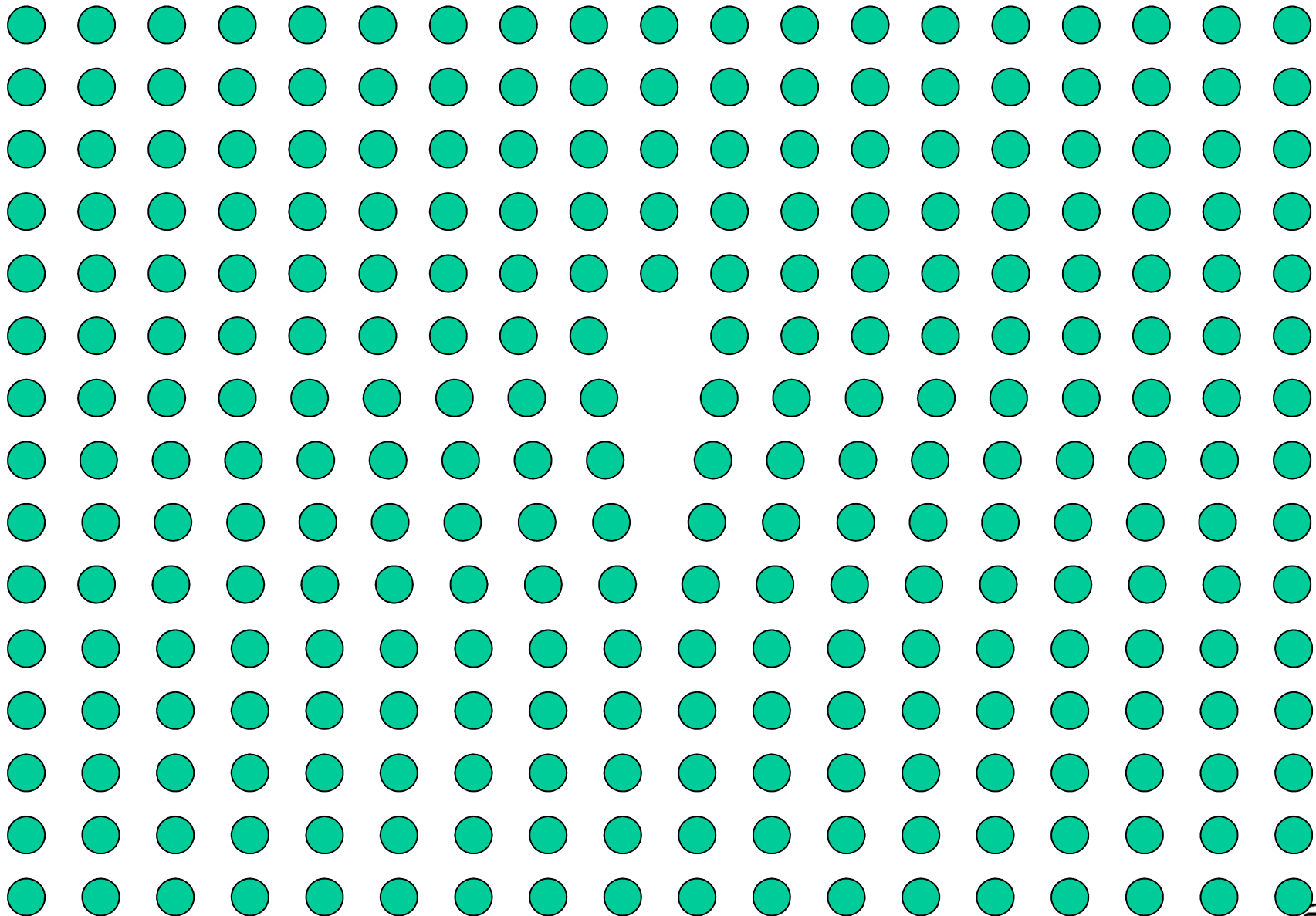
Dislocations



Missing half plane → A Defect



An extra half plane...



...or a missing half plane



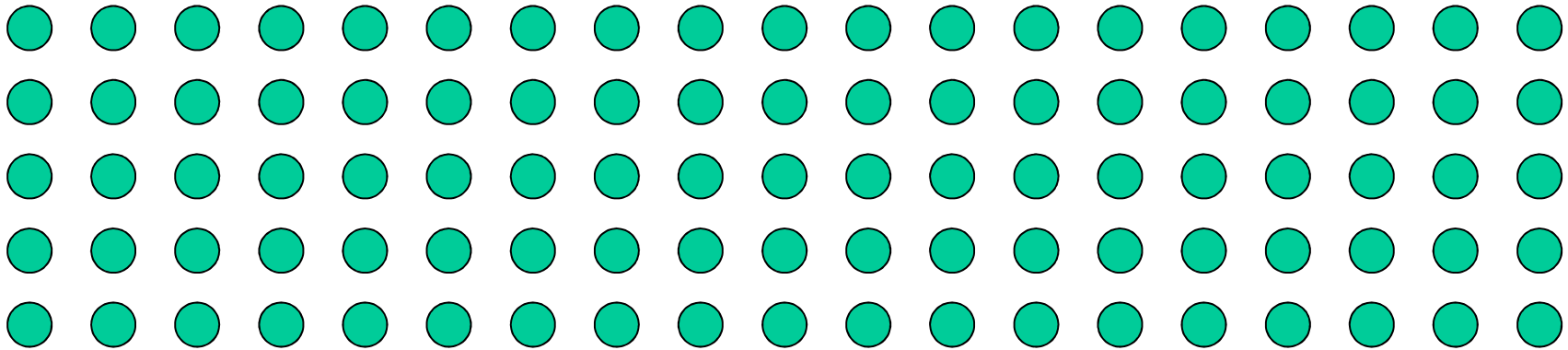
What kind of
defect is this?

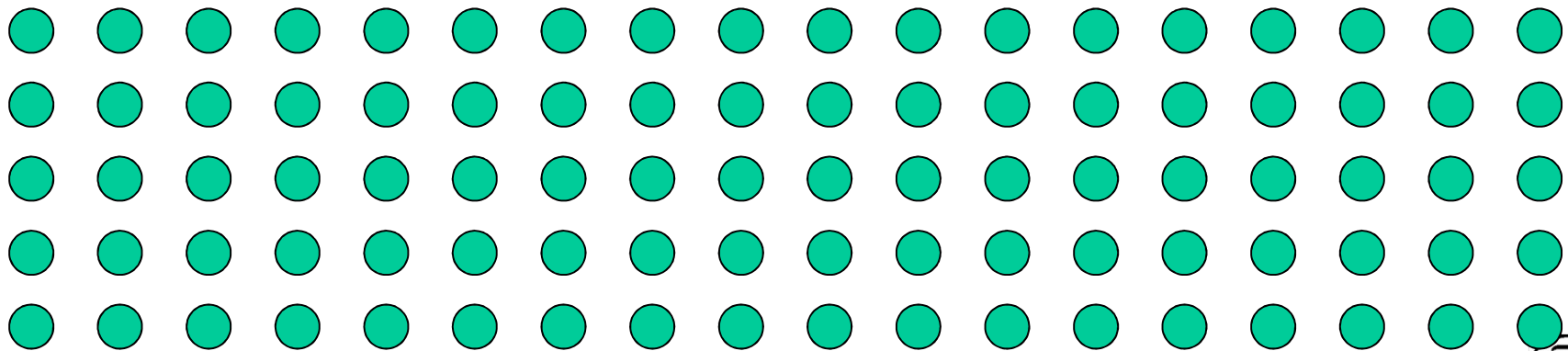
A line defect?

Or a planar
defect?



No extra plane!

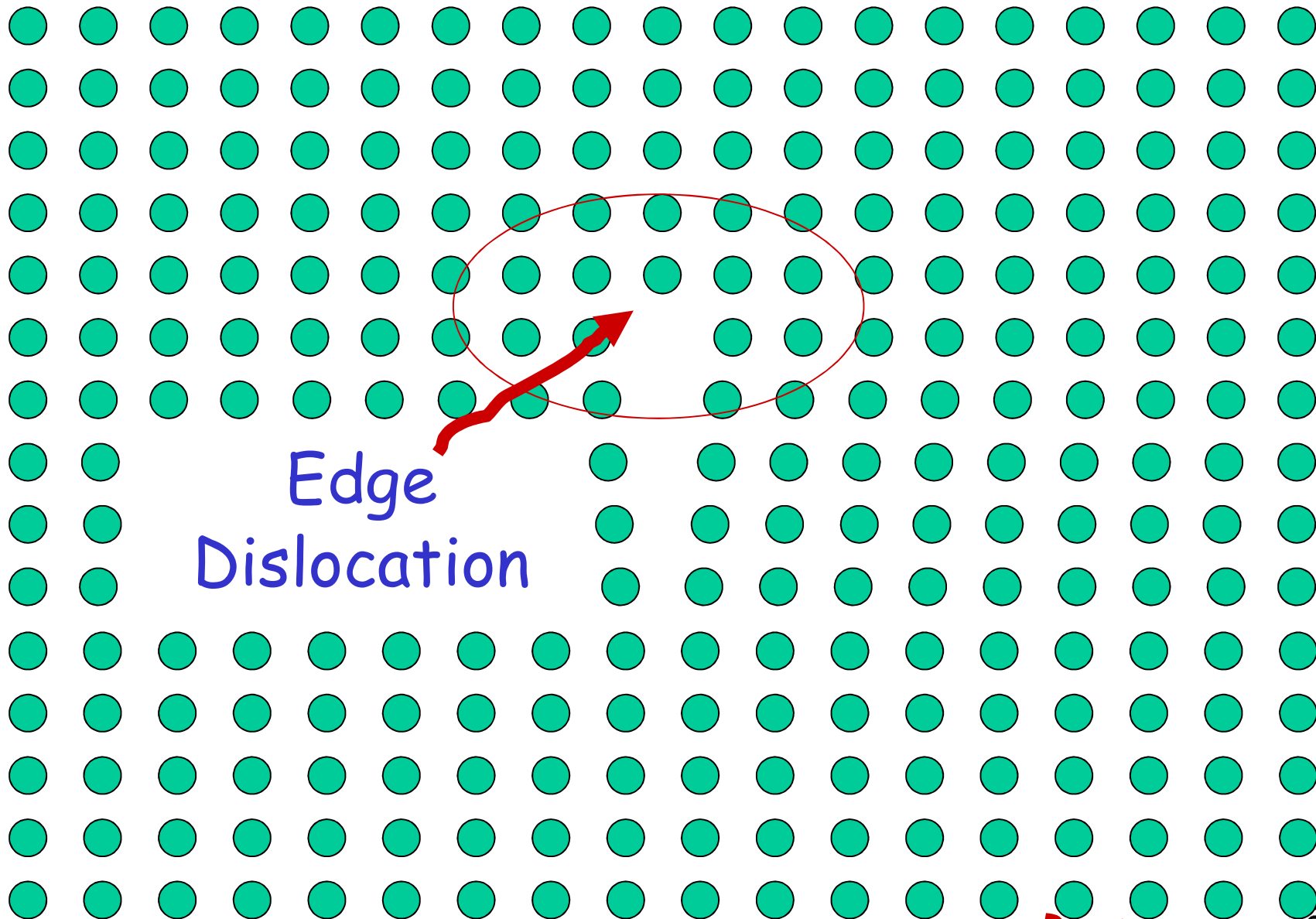




No missing plane!!!



An extra ~~half~~ plane...



Edge
Dislocation

...or a missing ~~half~~ plane



If a plane ends abruptly inside a crystal we have a defect.

The whole of abruptly ending plane is not a defect

Only the edge of the plane can be considered as a defect



This is a line defect called an
EDGE DISLOCATION

